GAUTAM KUMAR

Verification Engineer

gkt2work@gmail.com +91 9934455873 Bangalore, India

EXPERIENCE

Verification Engineer

Chipsil Technologies

March 2025 – Present Bangalore, India

- Functional verification of IP-level modules (e.g., FIFO, FSM, and AHB protocol) using SystemVerilog UVM
- Formal verification using SystemVerilog Assertions (SVA), including formal property verification, and connectivity verification
- Executed regressions and performed debugging to ensure functional and code coverage closure
- Created automation scripts for running regressions automatically

Intern - GenAI Applications

ForeignAdmits

December 2024 – July 2025 Bangalore, India

- Agentic-AI based chatbot development
- Used and developed Retrieval-Augmented Generation (RAG) based approaches to improve LLM outputs
- Connected user applications via the Model Context Protocol (MCP)

TRAININGS

ASIC Verification Course

Maven Silicon

2024

ASIC verification course on verification methodologies

VLSI for Beginners

National Institute of Electronics & Information Technology (NIELIT)

2023

Training on VLSI and its applications in the industry

EDUCATION

Executive M.Tech in VLSI Design

PES University (Maven Silicon)

July 2026 – June 2028 Bangalore, India

B.Tech in Electronics & Communication Engineering

Asansol Engineering College

August 2014 – June 2018 Asansol, India

PERSONAL PROJECTS

Open-Source Formal Verification Tool

Personal Project

2026 Bangalore, India

Tool portal

- Developed a complete GUI-based infrastructure to use SymbiYosys formal verification tool

Agentic-AI Chatbot for Verification

Personal Project

2025 Bangalore, India

GitHub repository

- Built an agentic-AI driven chatbot using RAG to generate RTL and testbench code automatically

Verification of ALU and FSM

Maven Silicon

2024 Bangalore, India

- Implemented a constrained-random SV-UVM test-bench architecture from scratch for functional verification of an ALU
- Created an SVA based formal verification suite and achieved coverage closure for FSM design

METHODOLOGIES

SV-UVM | Formal Verification

PROGRAMMING SKILLS

SystemVerilog
Verilog
VHDL
Python
TCL

EDA TOOLS

Xcelium | Jasper | vManager

AI/ML SKILLS

GenAI | LLM | Agentic-AI | RAG | MCP | Machine Learning

OTHER SKILLS

Linux | Jira | Bitbucket | Git | Github | GitLab | Confluence | VSCode | Eclipse